NC Grobner basis computation of NC polynomials annihilating the Arveson boundary of a free quadrilateral using NCGBX.

This version corrects an error in a GB reduction in the 1Sep20 version. Additionally, this version directly adjoins the inverse relations using the variable w rather than inv[c0+c1*z1+c2*z2]. The latter is not supported by NCAlgebra 6.0.3.

In[82]:=
```
<< NC`
<< NCAlgebra`
<< NCGBX`
RuleToPoly[rule_] := NCReplaceRepeated[rule, Rule → Subtract];
view[list_] := Substitute[list, NonCommutativeMultiply → Dot] // MatrixForm
NCReplaceRepeatedRepeated[polys_, ruls_, iters_] := Block[{iter, polysR},
  polysR = polys;
  For[iter = 0, iter < iters, iter++,
   polysR = NCE[NCReplaceRepeated[polysR, ruls]]];
  Return[polysR]]


SNC[w, h, z1, z2]
SetMonomialOrder[z1, z2, w, h]
Clear[c0, c1, c2]
```

NC: You are using the version of NCAlgebra which is found in: "C:\Users\Eric Work\NC\".

NCAlgebra: All lower cap single letter symbols (e.g. a,b,c,...) were set as noncommutative.

Out[89]= {{z1}, {z2}, {w}, {h}}

This notebook requires the NCAlgebra package available at http://math.ucsd.edu/~ncalg/.

Relevant paper: https://arxiv.org/abs/2008.13250

The generic case

General Poly Runs. With c0,c1,c2 unspecified. Here the relations generate all polys.

We compute a GB from the rational functions defining the Arveson boundary of our free quadrilateral. The GB contains nine NC rational functions, four of which are NC polynomials in z1,z2. Note that we have encoded that w is the inverse of (c0+c1*z1+c2*z2) via the relations 1-(c0+c1*z1+c2*z2)**w and 1-w**(c0+c1*z1+c2*z2).

Note: Since w = Inv[c0+c1*z1+c2*z2] functions containing w are understood to be rational functions.

In[11]:=
```
Polys = {c0 + c1 * z1 + c2 * z2 - z1 ** w ** z1, c0 + c1 * z1 + c2 * z2 - z2 ** w ** z2,
   1 - (c0 + c1 * z1 + c2 * z2) ** w, 1 - w ** (c0 + c1 * z1 + c2 * z2)}
```

Out[11]=
```
{c0 + c1 z1 + c2 z2 - z1 ** w ** z1, c0 + c1 z1 + c2 z2 - z2 ** w ** z2,
 1 - (c0 + c1 z1 + c2 z2) ** w, 1 - w ** (c0 + c1 z1 + c2 z2)}
```

In[12]:=
```
ColumnForm[Polys]
```

Out[12]=
```
c0 + c1 z1 + c2 z2 - z1 ** w ** z1
c0 + c1 z1 + c2 z2 - z2 ** w ** z2
1 - (c0 + c1 z1 + c2 z2) ** w
1 - w ** (c0 + c1 z1 + c2 z2)
```

In[13]:=
```
GeneralGB1 = NCMakeGB[Polys, 5] // FullSimplify
```

```
* * * * * * * * * * * * * * * * * *
* * *   NCPolyGroebner    * * *
* * * * * * * * * * * * * * * * * *
* Symbolic coefficients detected
* Monomial order:
> z1 ≪ z2 ≪ w ≪ h
* Reduce and normalize initial set
> Initial set could not be reduced
* Computing initial set of obstructions
> MAJOR Iteration 1, 7 polys in the basis, 11 obstructions
> MAJOR Iteration 2, 9 polys in the basis, 14 obstructions
* Found Groebner basis with 9 polynomials
* * * * * * * * * * * * * * * * * *
```

Out[13]=

$$\left\{z1 ** z2 ** z1 \to \frac{(c0 + c1\,z1)\left(c0^2 + 2\,c0\,c1\,z1 + \left(-1 + c1^2 - c2^2\right) z1^2\right) + c0^2\,c2\,z2 + c0\,c1\,c2\,(z1 ** z2 + z2 ** z1)}{-\left(\left(1 + c1^2\right) c2\right) + c2^3},\right.$$

$$z1 ** z2^2 \to \frac{1}{-\left(\left(1 + c1^2\right) c2\right) + c2^3}$$

$$\left(c0\left(-c2\,z1\,(c0 + c1\,z1) + c0\,c1\,z2 + c1\,c2\,z2^2 + 2\,(c1 - c2)\,(c1 + c2)\,z1 ** z2\right) + c1\left(-1 + c1^2 - c2^2\right) z1^2 ** z2\right), z2^2 ** z1 \to \frac{1}{-\left(\left(1 + c1^2\right) c2\right) + c2^3}$$

$$\left(c0\left(-c2\,z1\,(c0 + c1\,z1) + c0\,c1\,z2 + c1\,c2\,z2^2 + 2\,(c1 - c2)\,(c1 + c2)\,z2 ** z1\right) + c1\left(-1 + c1^2 - c2^2\right) z2 ** z1^2\right), z2^3 \to \frac{1}{-\left(\left(1 + c1^2\right) c2\right) + c2^3}$$

$$\left(-c0\left(c0\,(c0 + c1\,z1) + 3\,c0\,c2\,z2 - \left(1 + c1^2 - 3\,c2^2\right) z2^2 + c1\,c2\,(z1 ** z2 + z2 ** z1)\right) + c1\left(-1 + c1^2 - c2^2\right) z2 ** z1 ** z2\right),$$

$$w ** z1 \to \frac{c0\left(1 - c1^2 + c2^2\right) - c0^2\,w + c1\left(1 - c1^2 + c2^2\right) z1 + c2\left(-1 - c1^2 + c2^2\right) z2 - c0\,c1\,z1 ** w}{c0\,c1},$$

$$z1^2 ** w \to \frac{1}{c0\,c1}\left(-c0^2\,c1 + c0\left(1 - 2\,c1^2 + c2^2\right) z1 + c1\left(1 - c1^2 + c2^2\right) z1^2 - c0\,c1\,c2\,z2 - c0^2\,z1 ** w + c2\left(-1 - c1^2 + c2^2\right) z1 ** z2\right), z2 ** w \to -\frac{-1 + c0\,w + c1\,z1 ** w}{c2},$$

$$w ** z2 \to \frac{c0\,(c1 - c2)\,(c1 + c2) + c1\left(-1 + c1^2 - c2^2\right) z1 + c2\left(1 + c1^2 - c2^2\right) z2 + c0\,c1\,z1 ** w}{c0\,c2},$$

$$z2 ** z1 ** w \to \left.\frac{c0\,c2\,(c0 + c1\,z1) - c0\left(c1^2 - 2\,c2^2\right) z2 + c2\left(-1 - c1^2 + c2^2\right) z2^2 + c1\left(1 - c1^2 + c2^2\right) z2 ** z1}{c0\,c1}\right\}$$

We verify that the original functions Polys are contained in this ideal by reducing them to 0 via GeneralGB1.

In[15]:=
```
NCReplaceRepeatedRepeated[Polys, GeneralGB1, 10]
```

Out[15]=
```
{0, 0, 0, 0}
```

The first four rules in this GB correspond to polynomials in z1,z2.

```
GeneralGB1[[1 ;; 4]]
```

Out[16]=

$$\left\{z1 ** z2 ** z1 \to \frac{(c0 + c1\,z1)\left(c0^2 + 2\,c0\,c1\,z1 + \left(-1 + c1^2 - c2^2\right) z1^2\right) + c0^2\,c2\,z2 + c0\,c1\,c2\,(z1 ** z2 + z2 ** z1)}{-\left(\left(1 + c1^2\right) c2\right) + c2^3},\right.$$

$$z1 ** z2^2 \to \frac{1}{-\left(\left(1 + c1^2\right) c2\right) + c2^3}$$

$$\left(c0\left(-c2\,z1\,(c0 + c1\,z1) + c0\,c1\,z2 + c1\,c2\,z2^2 + 2\,(c1 - c2)\,(c1 + c2)\,z1 ** z2\right) + c1\left(-1 + c1^2 - c2^2\right) z1^2 ** z2\right), z2^2 ** z1 \to \frac{1}{-\left(\left(1 + c1^2\right) c2\right) + c2^3}$$

$$\left(c0\left(-c2\,z1\,(c0 + c1\,z1) + c0\,c1\,z2 + c1\,c2\,z2^2 + 2\,(c1 - c2)\,(c1 + c2)\,z2 ** z1\right) + c1\left(-1 + c1^2 - c2^2\right) z2 ** z1^2\right), z2^3 \to \frac{1}{-\left(\left(1 + c1^2\right) c2\right) + c2^3}$$

$$\left.\left(-c0\left(c0\,(c0 + c1\,z1) + 3\,c0\,c2\,z2 - \left(1 + c1^2 - 3\,c2^2\right) z2^2 + c1\,c2\,(z1 ** z2 + z2 ** z1)\right) + c1\left(-1 + c1^2 - c2^2\right) z2 ** z1 ** z2\right)\right\}$$

We will show that the polynomials in z1,z2 in GeneralGB1 above together with the invertibility relations generate the same ideal as GeneralGB1. I.e. if the polynomials in z1,z2 evaluate to zero, then all the rational functions in z1,z2,w evaluate to zero. To accomplish this we form a Grobner basis from these polynomials in z1,z2 together with the invertibility relations 1-(c0+c1*z1+c2*z2)**w,1-w**(c0+c1*z1+c2*z2) use it to reduce GeneralGB1

In[19]:=
```
GBpolys = RuleToPoly[
  Union[GeneralGB1[[1 ;; 4]], {1 - (c0 + c1 ** z1 + c2 ** z2) ** w, 1 - w ** (c0 + c1 ** z1 + c2 ** z2)}]]
PolysGB = NCMakeGB[GBpolys, 8];
```

Out[19]=
$$\left\{1 - w ** (c0 + c1\, z1 + c2\, z2),\ 1 - (c0 + c1\, z1 + c2\, z2) ** w,\ z2^3 - \frac{1}{-\left(\left(1 + c1^2\right) c2\right) + c2^3}\right.$$
$$\left(-c0\left(c0\,(c0 + c1\, z1) + 3\, c0\, c2\, z2 - \left(1 + c1^2 - 3\, c2^2\right) z2^2 + c1\, c2\,(z1 ** z2 + z2 ** z1)\right) + \right.$$
$$\left.c1\left(-1 + c1^2 - c2^2\right) z2 ** z1 ** z2\right),\ z1 ** z2^2 - \frac{1}{-\left(\left(1 + c1^2\right) c2\right) + c2^3}$$
$$\left(c0\left(-c2\, z1\,(c0 + c1\, z1) + c0\, c1\, z2 + c1\, c2\, z2^2 + 2\,(c1 - c2)\,(c1 + c2)\, z1 ** z2\right) + \right.$$
$$\left.c1\left(-1 + c1^2 - c2^2\right) z1^2 ** z2\right),\ -\frac{1}{-\left(\left(1 + c1^2\right) c2\right) + c2^3}$$
$$\left(c0\left(-c2\, z1\,(c0 + c1\, z1) + c0\, c1\, z2 + c1\, c2\, z2^2 + 2\,(c1 - c2)\,(c1 + c2)\, z2 ** z1\right) + \right.$$
$$\left.c1\left(-1 + c1^2 - c2^2\right) z2 ** z1^2\right) + z2^2 ** z1,$$
$$-\frac{(c0 + c1\, z1)\left(c0^2 + 2\, c0\, c1\, z1 + \left(-1 + c1^2 - c2^2\right) z1^2\right) + c0^2\, c2\, z2 + c0\, c1\, c2\,(z1 ** z2 + z2 ** z1)}{-\left(\left(1 + c1^2\right) c2\right) + c2^3} +$$
$$\left.z1 ** z2 ** z1\right\}$$

```
* * * * * * * * * * * * * * * * *
* * *   NCPolyGroebner    * * *
* * * * * * * * * * * * * * * * *
* Symbolic coefficients detected
* Monomial order:
> z1 ≪ z2 ≪ w ≪ h
* Reduce and normalize initial set
> Initial set could not be reduced
* Computing initial set of obstructions
> MAJOR Iteration 1, 10 polys in the basis, 25 obstructions
> MAJOR Iteration 2, 9 polys in the basis, 6 obstructions
* Found Groebner basis with 9 polynomials
* * * * * * * * * * * * * * * * *
```

In[21]:= **PolysGB**

Out[21]=
$$\left\{z1 ** z2 ** z1 \to \frac{c0^3}{c2\left(-1-c1^2+c2^2\right)}+\frac{3\,c0^2\,c1\,z1}{c2\left(-1-c1^2+c2^2\right)}-\frac{\left(c0-3\,c0\,c1^2+c0\,c2^2\right)z1^2}{c2\left(-1-c1^2+c2^2\right)}-\right.$$
$$\frac{\left(c1-c1^3+c1\,c2^2\right)z1^3}{c2\left(-1-c1^2+c2^2\right)}-\frac{c0^2\,z2}{1+c1^2-c2^2}-\frac{c0\,c1\,z1 ** z2}{1+c1^2-c2^2}-\frac{c0\,c1\,z2 ** z1}{1+c1^2-c2^2},$$
$$z1 ** z2^2 \to \frac{c0^2\,z1}{1+c1^2-c2^2}+\frac{c0\,c1\,z1^2}{1+c1^2-c2^2}+\frac{c0^2\,c1\,z2}{c2\left(-1-c1^2+c2^2\right)}-\frac{c0\,c1\,z2^2}{1+c1^2-c2^2}-$$
$$\frac{2\left(-c0\,c1^2+c0\,c2^2\right)z1 ** z2}{c2\left(-1-c1^2+c2^2\right)}-\frac{\left(c1-c1^3+c1\,c2^2\right)z1^2 ** z2}{c2\left(-1-c1^2+c2^2\right)},$$
$$z2^2 ** z1 \to \frac{c0^2\,z1}{1+c1^2-c2^2}+\frac{c0\,c1\,z1^2}{1+c1^2-c2^2}+\frac{c0^2\,c1\,z2}{c2\left(-1-c1^2+c2^2\right)}-\frac{c0\,c1\,z2^2}{1+c1^2-c2^2}-$$
$$\frac{2\left(-c0\,c1^2+c0\,c2^2\right)z2 ** z1}{c2\left(-1-c1^2+c2^2\right)}-\frac{\left(c1-c1^3+c1\,c2^2\right)z2 ** z1^2}{c2\left(-1-c1^2+c2^2\right)},$$
$$z2^3 \to -\frac{c0^3}{c2\left(-1-c1^2+c2^2\right)}-\frac{c0^2\,c1\,z1}{c2\left(-1-c1^2+c2^2\right)}+\frac{3\,c0^2\,z2}{1+c1^2-c2^2}-\frac{\left(-c0-c0\,c1^2+3\,c0\,c2^2\right)z2^2}{c2\left(-1-c1^2+c2^2\right)}+$$
$$\frac{c0\,c1\,z1 ** z2}{1+c1^2-c2^2}+\frac{c0\,c1\,z2 ** z1}{1+c1^2-c2^2}-\frac{\left(c1-c1^3+c1\,c2^2\right)z2 ** z1 ** z2}{c2\left(-1-c1^2+c2^2\right)},$$
$$w ** z1 \to -\frac{-1+c1^2-c2^2}{c1}-\frac{c0\,w}{c1}-\frac{\left(-1+c1^2-c2^2\right)z1}{c0}-\frac{c2\left(1+c1^2-c2^2\right)z2}{c0\,c1}-z1 ** w,$$
$$z1^2 ** w \to -c0-\frac{\left(-1+2\,c1^2-c2^2\right)z1}{c1}-\frac{\left(-1+c1^2-c2^2\right)z1^2}{c0}-c2\,z2-$$
$$\frac{c0\,z1 ** w}{c1}-\frac{c2\left(1+c1^2-c2^2\right)z1 ** z2}{c0\,c1},\ z2 ** w \to c2^{-1}-\frac{c0\,w}{c2}-\frac{c1\,z1 ** w}{c2},$$
$$w ** z2 \to c2^{-1}+\frac{-1+c1^2-c2^2}{c2}+\frac{c1\left(-1+c1^2-c2^2\right)z1}{c0\,c2}+\frac{\left(1+c1^2-c2^2\right)z2}{c0}+\frac{c1\,z1 ** w}{c2},$$
$$\left.z2 ** z1 ** w \to \frac{c0\,c2}{c1}+c2\,z1-\frac{\left(c1^2-2\,c2^2\right)z2}{c1}-\frac{c2\left(1+c1^2-c2^2\right)z2^2}{c0\,c1}-\frac{\left(-1+c1^2-c2^2\right)z2 ** z1}{c0}\right\}$$

This GB reduces all functions GeneralGB1. Similarly, the GeneralGB1 reduces all functions in PolysGB. We conclude the ideals are the same.

In[23]:=
```
GeneralPolys = RuleToPoly[GeneralGB1];
GeneralPolysSimplified =
 NCReplaceRepeatedRepeated[GeneralPolys, PolysGB, 10] // Expand // Simplify
PolysGBSimplified =
 NCReplaceRepeatedRepeated[RuleToPoly[PolysGB], GeneralGB1, 10] // Expand // Simplify
```

Out[24]= {0, 0, 0, 0, 0, 0, 0, 0, 0}

Out[25]= {0, 0, 0, 0, 0, 0, 0, 0, 0}

Special cases

We now consider several special cases of c0, c1, and c2 in the equations defining InvGB are not defined. Namely we consider the cases c1=0 and c2=0 and c1=c2=0 and $1 + c1^2 - c2^2 = 0$. As described in the main article, the c0=0 cannot occur, so we do not consider it here.

---

c1 = 0

In[26]:=
```
InvPolys = {c0 + c1 * z1 + c2 * z2 - z1 ** w ** z1, c0 + c1 * z1 + c2 * z2 - z2 ** w ** z2,
    1 - (c0 + c1 * z1 + c2 * z2) ** w, 1 - w ** (c0 + c1 * z1 + c2 * z2)} //. {c1 → 0};
SetMonomialOrder[z1, z2, w]
InvGB = NCMakeGB[InvPolys, 4];
```

Out[27]= {{z1}, {z2}, {w}}

```
* * * * * * * * * * * * * * * * *
* * *   NCPolyGroebner    * * *
* * * * * * * * * * * * * * * * *
* Symbolic coefficients detected
* Monomial order:
> z1 ≪ z2 ≪ w
* Reduce and normalize initial set
> Initial set reduced to '3' out of '4' polynomials
* Computing initial set of obstructions
* Found Groebner basis with 3 polynomials
* * * * * * * * * * * * * * * * *
```

In this case we obtain 3 equation two of which are polynomials

In[30]:=
```
InvGB
```

Out[30]=
$$\left\{z1 ** z2 ** z1 \to \frac{c0^3}{c2\left(-1+c2^2\right)} - \frac{c0\left(1+c2^2\right) z1^2}{c2\left(-1+c2^2\right)} + \frac{c0^2\, z2}{-1+c2^2},\right.$$
$$\left. z2^2 \to -\frac{c0^2}{-1+c2^2} - \frac{2\, c0\, c2\, z2}{-1+c2^2},\ w \to -\frac{-1-c2^2}{c0} + \frac{c2\left(-1+c2^2\right) z2}{c0^2}\right\}$$

In[31]:=
```
GBpolys = {InvGB[[1]], InvGB[[2]]}
```

Out[31]=
$$\left\{z1 ** z2 ** z1 \to \frac{c0^3}{c2\left(-1+c2^2\right)} - \frac{c0\left(1+c2^2\right) z1^2}{c2\left(-1+c2^2\right)} + \frac{c0^2\, z2}{-1+c2^2},\ z2^2 \to -\frac{c0^2}{-1+c2^2} - \frac{2\, c0\, c2\, z2}{-1+c2^2}\right\}$$

Since c1 = 0, (c0+c2*z2) is the inverse of w. From this we see that the rational equation is reduced by the polynomials. In particular, (c0 +c2 z2) is the inverse of w and after reducing by the polynomials RuleToPoly[InvGB[[3]]]**(c0+c2 z2) = -1+c0 w+c2 w**z2 = -1 + w**(c0 + c2 z2) = 0. Thus RuleToPoly[InvGB[[3]]]**(c0+c2 z2) is in the ideal. Using invertibility of c0+c2 z2 gives that RuleToPoly[InvGB[[3]]] is also in the ideal.

In[33]:=
```
InvGB[[3]]
NCReplaceRepeated[NCExpand[RuleToPoly[InvGB[[3]]] ** (c0 + c2 z2)], GBpolys] //
 FullSimplify
```

Out[33]=
$$w \to -\frac{-1-c2^2}{c0} + \frac{c2\left(-1+c2^2\right) z2}{c0^2}$$

Out[34]=
$$-1 + c0\, w + c2\, w ** z2$$

As subcases of c1 = 0, we also consider the case where c2^2 = 1, i.e., c2 = 1 or -1. We first consider c2 = 1.

In[35]:=
```
InvPolys = {c0 + c1 * z1 + c2 * z2 - z1 ** w ** z1, c0 + c1 * z1 + c2 * z2 - z2 ** w ** z2,
    1 - (c0 + c1 * z1 + c2 * z2) ** w, 1 - w ** (c0 + c1 * z1 + c2 * z2)} //. {c1 → 0, c2 → 1};
SetMonomialOrder[z1, z2, w]
InvGB = NCMakeGB[InvPolys, 4];
```

Out[36]=
```
{{z1}, {z2}, {w}}
```

```
* * * * * * * * * * * * * * * * * *
* * *   NCPolyGroebner    * * *
* * * * * * * * * * * * * * * * * *
* Symbolic coefficients detected
* Monomial order:
> z1 ≪ z2 ≪ w
* Reduce and normalize initial set
> Initial set reduced to '3' out of '4' polynomials
* Computing initial set of obstructions
* Found Groebner basis with 3 polynomials
* * * * * * * * * * * * * * * * * *
```

In this case w is a constant multiple of the identity and so all equations in the GB are polynomials in z1,z2.

In[38]:=
```
InvGB
```

Out[38]= $\left\{z1^2 \to \frac{c0^2}{4}, z2 \to -\frac{c0}{2}, w \to \frac{2}{c0}\right\}$

For completeness we include c2 = -1, which is similar to the above case.

In[39]:=
```
InvPolys = {c0 + c1 * z1 + c2 * z2 - z1 ** w ** z1, c0 + c1 * z1 + c2 * z2 - z2 ** w ** z2,
    1 - (c0 + c1 * z1 + c2 * z2) ** w, 1 - w ** (c0 + c1 * z1 + c2 * z2)} //. {c1 → 0, c2 → -1};
SetMonomialOrder[z1, z2, w]
InvGB = NCMakeGB[InvPolys, 4];
```

Out[40]=
```
{{z1}, {z2}, {w}}
```

```
* * * * * * * * * * * * * * * * * *
* * *   NCPolyGroebner    * * *
* * * * * * * * * * * * * * * * * *
* Symbolic coefficients detected
* Monomial order:
> z1 ≪ z2 ≪ w
* Reduce and normalize initial set
> Initial set reduced to '3' out of '4' polynomials
* Computing initial set of obstructions
* Found Groebner basis with 3 polynomials
* * * * * * * * * * * * * * * * * *
```

In this case w is a constant multiple of the identity and so all equations in the GB are polynomials in z1,z2.

In[42]:=
```
InvGB
```

Out[42]=
$$\left\{z1^2 \to \frac{c0^2}{4},\ z2 \to \frac{c0}{2},\ w \to \frac{2}{c0}\right\}$$

---

c2 = 0

In[43]:=
```
InvPolys = {c0 + c1 * z1 + c2 * z2 - z1 ** w ** z1, c0 + c1 * z1 + c2 * z2 - z2 ** w ** z2,
    1 - (c0 + c1 * z1 + c2 * z2) ** w, 1 - w ** (c0 + c1 * z1 + c2 * z2)} //. {c2 → 0};
SetMonomialOrder[z1, z2, w]
InvGB = NCMakeGB[InvPolys, 4];
```

Out[44]=
```
{{z1}, {z2}, {w}}
```

```
* * * * * * * * * * * * * * * * *
* * *   NCPolyGroebner    * * *
* * * * * * * * * * * * * * * * *
* Symbolic coefficients detected
* Monomial order:
> z1 ≪ z2 ≪ w
* Reduce and normalize initial set
> Initial set reduced to '3' out of '4' polynomials
* Computing initial set of obstructions
* Found Groebner basis with 3 polynomials
* * * * * * * * * * * * * * * * *
```

In this case we obtain 3 equation two of which are polynomials. Similar to the c1 =0 case, we use the invertibility of (c0+c1*z1) to conclude that the rational equation is reduced by the polynomials.

In[47]:=
```
InvGB
```

Out[47]=
$$\left\{z1^2 \to -\frac{c0^2}{-1+c1^2} - \frac{2\,c0\,c1\,z1}{-1+c1^2},\right.$$
$$\left. z2 ** z1 ** z2 \to \frac{c0^3}{c1\left(-1+c1^2\right)} + \frac{c0^2\,z1}{-1+c1^2} - \frac{c0\left(1+c1^2\right)z2^2}{c1\left(-1+c1^2\right)},\ w \to -\frac{-1-c1^2}{c0} + \frac{c1\left(-1+c1^2\right)z1}{c0^2}\right\}$$

In[48]:=
```
GBpolys = {InvGB[[1]], InvGB[[2]]}
```

Out[48]=
$$\left\{z1^2 \to -\frac{c0^2}{-1+c1^2} - \frac{2\,c0\,c1\,z1}{-1+c1^2},\ z2 ** z1 ** z2 \to \frac{c0^3}{c1\left(-1+c1^2\right)} + \frac{c0^2\,z1}{-1+c1^2} - \frac{c0\left(1+c1^2\right)z2^2}{c1\left(-1+c1^2\right)}\right\}$$

Since c2 = 0, (c0+c1*z1) is the inverse of w. From this we see that the rational equation is reduced by the polynomials.

In[50]:=
```
InvGB[[3]]
NCReplaceRepeated[NCExpand[RuleToPoly[InvGB[[3]]] ** (c0 + c1 z1)], GBpolys] //
 FullSimplify
```

Out[50]= $$w \to -\frac{-1-c1^2}{c0}+\frac{c1\left(-1+c1^2\right) z1}{c0^2}$$

Out[51]= $$-1+c0\,w+c1\,w ** z1$$

The c1^2 =1 subcases are similar to the above and are included for completeness. We first consider c1 = 1.

In[52]:=
```
InvPolys = {c0 + c1 * z1 + c2 * z2 - z1 ** w ** z1, c0 + c1 * z1 + c2 * z2 - z2 ** w ** z2,
    1 - (c0 + c1 * z1 + c2 * z2) ** w, 1 - w ** (c0 + c1 * z1 + c2 * z2)} //. {c2 → 0, c1 → 1};
SetMonomialOrder[z1, z2, w]
InvGB = NCMakeGB[InvPolys, 4];
```

Out[53]=
```
{{z1}, {z2}, {w}}
```

```
* * * * * * * * * * * * * * * * *
* * *   NCPolyGroebner    * * *
* * * * * * * * * * * * * * * * *
* Symbolic coefficients detected
* Monomial order:
> z1 ≪ z2 ≪ w
* Reduce and normalize initial set
> Initial set reduced to '3' out of '4' polynomials
* Computing initial set of obstructions
* Found Groebner basis with 3 polynomials
* * * * * * * * * * * * * * * * *
```

In this case w is a constant multiple of the identity and so all equations in the GB are polynomials in z1,z2.

In[55]:=
```
InvGB
```

Out[55]= $$\left\{z1 \to -\frac{c0}{2},\ z2^2 \to \frac{c0^2}{4},\ w \to \frac{2}{c0}\right\}$$

We now consider c1 = -1

In[56]:=
```
InvPolys = {c0 + c1 * z1 + c2 * z2 - z1 ** w ** z1, c0 + c1 * z1 + c2 * z2 - z2 ** w ** z2,
    1 - (c0 + c1 * z1 + c2 * z2) ** w, 1 - w ** (c0 + c1 * z1 + c2 * z2)} //. {c2 → 0, c1 → -1};
SetMonomialOrder[z1, z2, w]
InvGB = NCMakeGB[InvPolys, 4];
```

Out[57]=
```
{{z1}, {z2}, {w}}
```

```
* * * * * * * * * * * * * * * * * *
* * *   NCPolyGroebner    * * *
* * * * * * * * * * * * * * * * * *
* Symbolic coefficients detected
* Monomial order:
> z1 ≪ z2 ≪ w
* Reduce and normalize initial set
> Initial set reduced to '3' out of '4' polynomials
* Computing initial set of obstructions
* Found Groebner basis with 3 polynomials
* * * * * * * * * * * * * * * * * *
```

In this case w is a constant multiple of the identity and so all equations in the GB are polynomials in z1,z2.

In[59]:=
```
InvGB
```

Out[59]= $\left\{z1 \to \frac{c0}{2},\ z2^2 \to \frac{c0^2}{4},\ w \to \frac{2}{c0}\right\}$

---

c1 = c2 = 0

In[60]:=
```
InvPolys = {c0 + c1 * z1 + c2 * z2 - z1 ** w ** z1, c0 + c1 * z1 + c2 * z2 - z2 ** w ** z2,
    1 - (c0 + c1 * z1 + c2 * z2) ** w, 1 - w ** (c0 + c1 * z1 + c2 * z2)} //. {c1 → 0, c2 -> 0};
InvPolys // ColumnForm
```

Out[61]=
```
c0 - z1 ** w ** z1
c0 - z2 ** w ** z2
1 - c0 w
1 - c0 w
```

In this case 1-c0 w = 0 and so w is a constant multiple of the identity. Thus all functions are polynomials in z1 and z2.

$1 + c1^2 - c2^2 = 0$ and $c1 \neq 0$.

Here we consider two subcases.

$$c2 = -\sqrt{1 + c1^2}$$

In[63]:=
```
InvPolys = {c0 + c1 * z1 + c2 * z2 - z1 ** w ** z1, c0 + c1 * z1 + c2 * z2 - z2 ** w ** z2,
    1 - (c0 + c1 * z1 + c2 * z2) ** w, 1 - w ** (c0 + c1 * z1 + c2 * z2)} //. {c2 → -Sqrt[1 + c1^2]};
SetMonomialOrder[z1, z2, w]
InvGB = NCMakeGB[InvPolys, 4];
```

Out[64]=
```
{{z1}, {z2}, {w}}

* * * * * * * * * * * * * * * * *

* * *   NCPolyGroebner    * * *

* * * * * * * * * * * * * * * * *

* Symbolic coefficients detected

* Monomial order:

> z1 ≪ z2 ≪ w

* Reduce and normalize initial set

> Initial set could not be reduced

* Computing initial set of obstructions

> MAJOR Iteration 1, 6 polys in the basis, 6 obstructions

* Found Groebner basis with 6 polynomials

* * * * * * * * * * * * * * * * *
```

We have two polynomials in z1,z2

In[67]:=
```
InvGB // ColumnForm
```

Out[67]=

$$z2 ** z1 \to \frac{c0^2}{c1\sqrt{1+c1^2}} + \frac{3\,c0\,z1}{\sqrt{1+c1^2}} + \frac{2\left(-1+c1^2\right)z1^2}{c1\sqrt{1+c1^2}} - \frac{2\,z1^3}{c0\sqrt{1+c1^2}} - \frac{c0\,z2}{c1} - z1 ** z2$$

$$z2^2 \to \left(\frac{3\,c0\,c1}{1+c1^2} - \frac{-c0+2\,c0\,c1^2}{c1\left(1+c1^2\right)}\right)z1 + \left(-\frac{-3+c1^2}{1+c1^2} + \frac{2\left(-1+c1^2\right)}{1+c1^2}\right)z1^2 + \frac{c0\,z2}{\sqrt{1+c1^2}} + \left(-\frac{c1}{\sqrt{1+c1^2}} - \frac{2-c1^2}{c1\sqrt{1+c1^2}}\right)z1 ** z2 - \frac{2\,z1^2 ** z}{c0\sqrt{1+c}}$$

$$w ** z1 \to \frac{2}{c1} - \frac{c0\,w}{c1} + \frac{2\,z1}{c0} - z1 ** w$$

$$z1^2 ** w \to -c0 - \frac{\left(-2+c1^2\right)z1}{c1} + \frac{2\,z1^2}{c0} + \sqrt{1+c1^2}\,z2 - \frac{c0\,z1 ** w}{c1}$$

$$z2 ** w \to -\frac{1}{\sqrt{1+c1^2}} + \frac{c0\,w}{\sqrt{1+c1^2}} + \frac{c1\,z1 ** w}{\sqrt{1+c1^2}}$$

$$w ** z2 \to \frac{1}{\sqrt{1+c1^2}} + \frac{2\,c1\,z1}{c0\sqrt{1+c1^2}} - \frac{c1\,z1 ** w}{\sqrt{1+c1^2}}$$

As in the generic case we form a Grobner basis using the polynomials in z1 and z2 together with the inverse relations reduces the remaining rational equations.

In[69]:=
```
GBpolys = Join[InvGB[[1 ;; 2]],
    {1 - (c0 + c1 * z1 + c2 * z2) ** w, 1 - w ** (c0 + c1 * z1 + c2 * z2)}] //. {c2 → -Sqrt[1 + c1^2]};
PolysGB = NCMakeGB[GBpolys, 4];
InvPolys = Map[RuleToPoly, InvGB];
InvGBSimplified = NCReplaceRepeated[InvPolys, PolysGB] // Simplify
```

```
* * * * * * * * * * * * * * * * * *
* * *   NCPolyGroebner    * * *
* * * * * * * * * * * * * * * * * *
* Symbolic coefficients detected
* Monomial order:
> z1 ≪ z2 ≪ w
* Reduce and normalize initial set
> Initial set could not be reduced
* Computing initial set of obstructions
> MAJOR Iteration 1, 7 polys in the basis, 12 obstructions
> MAJOR Iteration 2, 6 polys in the basis, 3 obstructions
* Found Groebner basis with 6 polynomials
* * * * * * * * * * * * * * * * * *
```

Out[72]= {0, 0, 0, 0, 0, 0}

$$c2 = \sqrt{1 + c1^2}$$

In[73]:=
```
InvPolys = {c0 + c1 * z1 + c2 * z2 - z1 ** w ** z1, c0 + c1 * z1 + c2 * z2 - z2 ** w ** z2,
    1 - (c0 + c1 * z1 + c2 * z2) ** w, 1 - w ** (c0 + c1 * z1 + c2 * z2)} //. {c2 → Sqrt[1 + c1^2]};
SetMonomialOrder[z1, z2, w]
InvGB = NCMakeGB[InvPolys, 4];
```

Out[74]= {{z1}, {z2}, {w}}

```
* * * * * * * * * * * * * * * * * *
* * *   NCPolyGroebner    * * *
* * * * * * * * * * * * * * * * * *
* Symbolic coefficients detected
* Monomial order:
> z1 ≪ z2 ≪ w
* Reduce and normalize initial set
> Initial set could not be reduced
* Computing initial set of obstructions
> MAJOR Iteration 1, 6 polys in the basis, 6 obstructions
* Found Groebner basis with 6 polynomials
* * * * * * * * * * * * * * * * * *
```

In[76]:=
```
InvGB // ColumnForm
```

Out[76]=

$$z2 ** z1 \to -\frac{c0^2}{c1\sqrt{1+c1^2}} - \frac{3\,c0\,z1}{\sqrt{1+c1^2}} - \frac{2\left(-1+c1^2\right)z1^2}{c1\sqrt{1+c1^2}} + \frac{2\,z1^3}{c0\sqrt{1+c1^2}} - \frac{c0\,z2}{c1} - z1 ** z2$$

$$z2^2 \to \left(\frac{3\,c0\,c1}{1+c1^2} - \frac{-c0+2\,c0\,c1^2}{c1\left(1+c1^2\right)}\right) z1 + \left(-\frac{-3+c1^2}{1+c1^2} + \frac{2\left(-1+c1^2\right)}{1+c1^2}\right) z1^2 - \frac{c0\,z2}{\sqrt{1+c1^2}} + \left(\frac{c1}{\sqrt{1+c1^2}} - \frac{-2+c1^2}{c1\sqrt{1+c1^2}}\right) z1 ** z2 + \frac{2\,z1^2 ** z2}{c0\sqrt{1+c1^2}}$$

$$w ** z1 \to \frac{2}{c1} - \frac{c0\,w}{c1} + \frac{2\,z1}{c0} - z1 ** w$$

$$z1^2 ** w \to -c0 - \frac{\left(-2+c1^2\right)z1}{c1} + \frac{2\,z1^2}{c0} - \sqrt{1+c1^2}\,z2 - \frac{c0\,z1 ** w}{c1}$$

$$z2 ** w \to \frac{1}{\sqrt{1+c1^2}} - \frac{c0\,w}{\sqrt{1+c1^2}} - \frac{c1\,z1 ** w}{\sqrt{1+c1^2}}$$

$$w ** z2 \to -\frac{1}{\sqrt{1+c1^2}} - \frac{2\,c1\,z1}{c0\sqrt{1+c1^2}} + \frac{c1\,z1 ** w}{\sqrt{1+c1^2}}$$

As in the generic case we form a Grobner basis using the polynomials in z1 and z2 together with the inverse relations reduces the remaining rational equations.

In[78]:=
```
GBpolys = Join[InvGB[[1 ;; 2]],
    {1 - (c0 + c1 * z1 + c2 * z2) ** w, 1 - w ** (c0 + c1 * z1 + c2 * z2)}] //. {c2 → Sqrt[1 + c1^2]};;
PolysGB = NCMakeGB[GBpolys, 4];
InvPolys = Map[RuleToPoly, InvGB];
InvGBSimplified = NCReplaceRepeated[InvPolys, PolysGB] // Simplify
```

```
* * * * * * * * * * * * * * * * * *
* * *   NCPolyGroebner    * * *
* * * * * * * * * * * * * * * * * *
* Symbolic coefficients detected
* Monomial order:
> z1 ≪ z2 ≪ w
* Reduce and normalize initial set
> Initial set could not be reduced
* Computing initial set of obstructions
> MAJOR Iteration 1, 7 polys in the basis, 12 obstructions
> MAJOR Iteration 2, 6 polys in the basis, 3 obstructions
* Found Groebner basis with 6 polynomials
* * * * * * * * * * * * * * * * * *
```

Out[81]= {0, 0, 0, 0, 0, 0}